

descriptives

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```
starwars <- dplyr::starwars
```

```
names(starwars)
```

```
[1] "name"      "height"    "mass"      "hair_color" "skin_color"  
[6] "eye_color" "birth_year" "sex"       "gender"     "homeworld"  
[11] "species"   "films"     "vehicles"  "starships"
```

```
glimpse(starwars)
```

```
Rows: 87
```

```
Columns: 14
```

```
$ name      <chr> "Luke Skywalker", "C-3PO", "R2-D2", "Darth Vader", "Leia Or~  
$ height    <int> 172, 167, 96, 202, 150, 178, 165, 97, 183, 182, 188, 180, 2~  
$ mass      <dbl> 77.0, 75.0, 32.0, 136.0, 49.0, 120.0, 75.0, 32.0, 84.0, 77.~  
$ hair_color <chr> "blond", NA, NA, "none", "brown", "brown, grey", "brown", N~  
$ skin_color <chr> "fair", "gold", "white, blue", "white", "light", "light", "~  
$ eye_color  <chr> "blue", "yellow", "red", "yellow", "brown", "blue", "blue",~  
$ birth_year <dbl> 19.0, 112.0, 33.0, 41.9, 19.0, 52.0, 47.0, NA, 24.0, 57.0, ~  
$ sex        <chr> "male", "none", "none", "male", "female", "male", "female",~  
$ gender     <chr> "masculine", "masculine", "masculine", "masculine", "femini~  
$ homeworld  <chr> "Tatooine", "Tatooine", "Naboo", "Tatooine", "Alderaan", "T~  
$ species    <chr> "Human", "Droid", "Droid", "Human", "Human", "Human", "Huma~  
$ films      <list> <"A New Hope", "The Empire Strikes Back", "Return of the J~  
$ vehicles   <list> <"Snowspeeder", "Imperial Speeder Bike">, <>, <>, <>, "Imp~  
$ starships  <list> <"X-wing", "Imperial shuttle">, <>, <>, "TIE Advanced x1",~
```

1 Descriptive Statistics

<u>Population</u>	<u>Sample</u>
<u>Parameters</u>	<u>Statistics</u>

1.1 Location / Central Tendency

1.1.1 Mode

The mode of a sample is very simple: it is the value that occurs **most frequently**.

A frequency table could be examined.

```
descr::freq(as.ordered(starwars$eye_color), plot = FALSE)
```

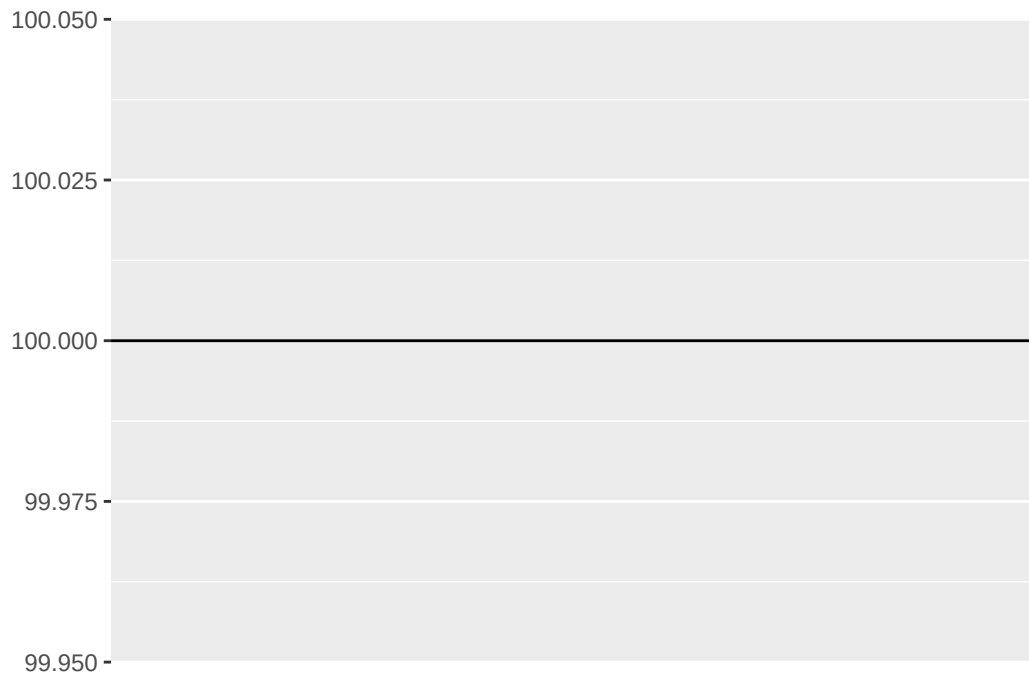
```
as.ordered(starwars$eye_color)
      Frequency Percent Cum Percent
black          10  11.494      11.49
blue           19  21.839      33.33
blue-gray       1   1.149      34.48
brown          21  24.138      58.62
dark            1   1.149      59.77
gold            1   1.149      60.92
green, yellow   1   1.149      62.07
hazel           3   3.448      65.52
orange          8   9.195      74.71
pink            1   1.149      75.86
red             5   5.747      81.61
red, blue       1   1.149      82.76
unknown         3   3.448      86.21
white           1   1.149      87.36
yellow          11  12.644     100.00
Total          87 100.000
```

```
modeOf(x = starwars$eye_color)
```

```
[1] "brown"
```

1.1.1.1 No Mode = same frequency

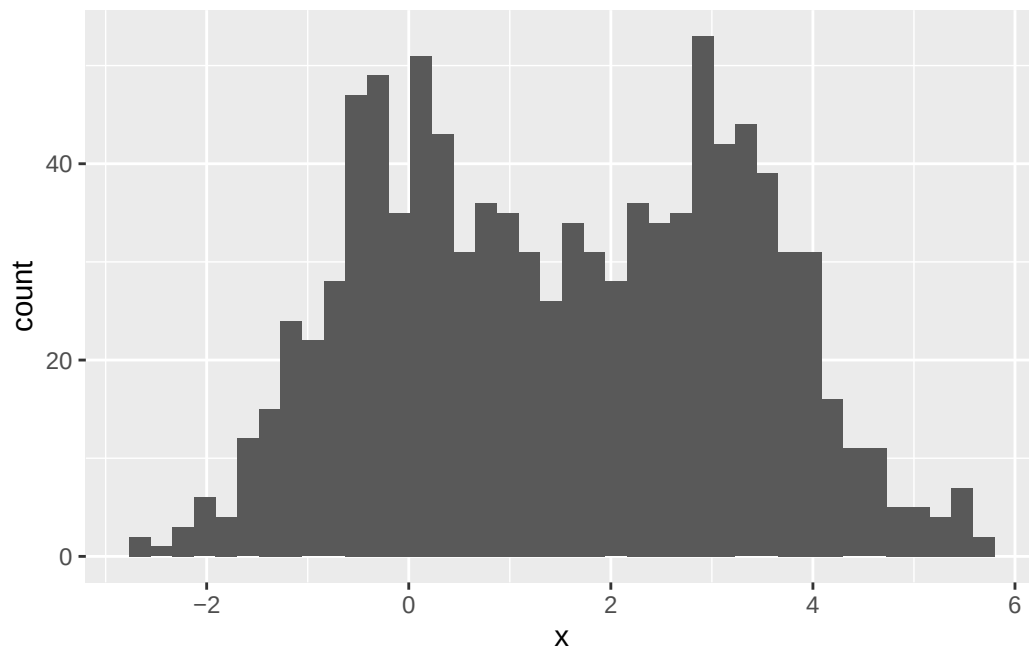
```
y = 100  
  
ggplot() +  
  geom_hline(yintercept = y)
```



1.1.1.2 Bimodal = two modes

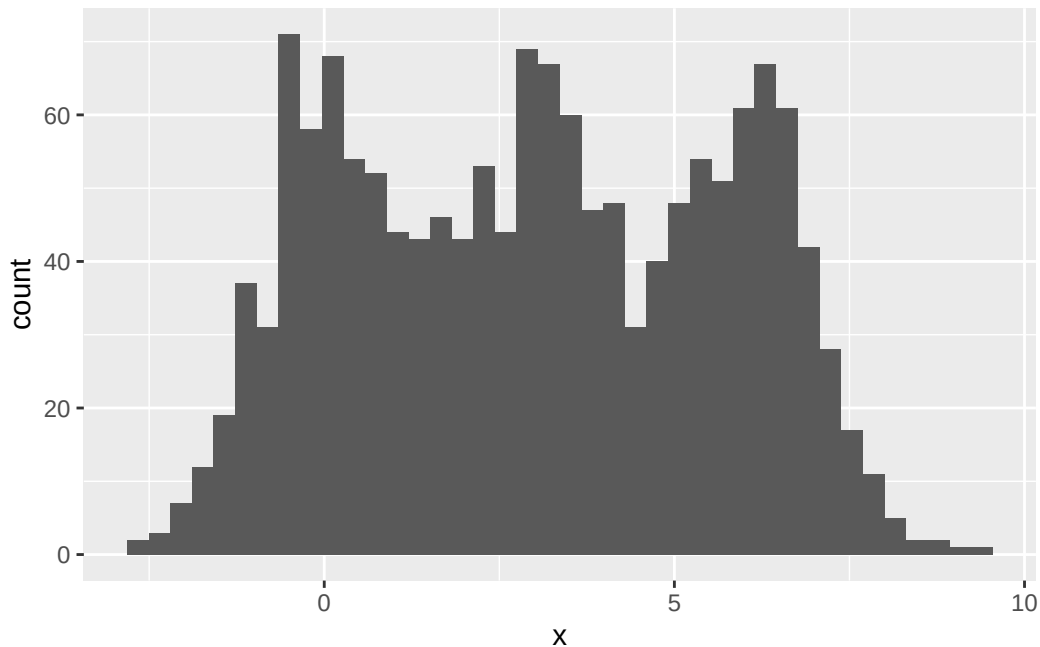
```
# Create data  
set.seed(123)  
  
n <- 1000 # Sample split in each below  
  
x <- c(  
  rnorm(n / 2, mean = 0, sd = 1), # Update mean or sd  
  rnorm(n / 2, mean = 3, sd = 1)  
)  
  
df <- data.frame(x)  
  
# Visualize
```

```
ggplot(df, aes(x)) +  
  geom_histogram(bins = 40)
```



1.1.1.3 Multimodal = >2 modes

```
# Create data  
set.seed(123)  
  
n <- 1500 # Sample split in each below  
  
x <- c(  
  rnorm(n / 3, mean = 0, sd = 1), # Update mean or sd  
  rnorm(n / 3, mean = 3, sd = 1),  
  rnorm(n / 3, mean = 6, sd = 1)  
)  
  
df <- data.frame(x)  
  
# Visualize  
ggplot(df, aes(x)) +  
  geom_histogram(bins = 40)
```



1.1.2 Median

The median of a set of observations is just the **middle value**. Let's imagine we were interested only in the first 5 AFL winning margins. To figure out the median, we sort these numbers into ascending order:

8, 31, **32**, 56, 56

From inspection, it's obvious that the median value of these 5 observations is 32, since that's the middle one in the sorted list (I've put it in bold to make it even more obvious). Easy stuff.

But what should we do if we were interested in the first 6 games rather than the first 5? Since the sixth game in the season had a winning margin of 14 points, our sorted list is now.

8, 14, **31**, **32**, 56, 56

and there are *two* middle numbers, 31 and 32. The median is defined as the average of those two numbers, which is of course 31.5.

As before, it's very tedious to do this by hand when you've got lots of numbers. In real life, of course, no-one actually calculates the median by sorting the data and then looking for the middle value. In real life, we use the median command:

```
median(starwars$mass, na.rm = TRUE)
```

```
[1] 79
```

```
# If you variable contains even one NA, this function will return NA, unless you tell it to .
```

1.1.3 Arithmetic Mean

Typical, old-fashioned average. Add all of the values up, and then divide by the total number of values.

$$\frac{1}{n} (x_1 + x_2 + \dots + x_n)$$

where $x_i, x_2 \dots x_n$ are the observed values and n is the number of observations.

Population	Sample
$\mu = \frac{\sum y}{N}$	$m = \frac{\sum y}{n}$

For example, for the values 56, 31, 56, 8, 32:

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

$$\bar{x} = \frac{56 + 31 + 56 + 8 + 32}{5}$$

$$\bar{x} = \frac{183}{5} = 36.60$$

Written out:

```
(56 + 31 + 56 + 8 + 32) / 5
```

```
[1] 36.6
```

We can tell R to same thing with functions.